Phase 2: Fundamental Features of Machine Learning on Azure

1. Azure Machine Learning Workspace:

Azure ML Workspace: The top-level resource for Azure Machine Learning, providing a centralized place to work with all the artifacts (data, compute, models, pipelines) you create.

2. Three Ways to Build Models in Azure ML:

A. Automated ML (AutoML)

A feature that automates the process of applying machine learning algorithms to data, automatically selecting the best algorithm and hyperparameters.

B. Azure ML Designer

A visual drag-and-drop interface used to build, test, and deploy machine learning models using pre-built modules without writing code.

C. Notebooks

An integrated development environment (IDE) inside Azure ML where data scientists write Python/R code directly using Jupyter Notebooks.

Crucial Exam Topic: Core Compute Resources

1. Compute Instances: Development workstations for data scientists, fully configured with Jupyter and cloud-based coding environments.

2. Compute Clusters: Scalable clusters of virtual machines used for heavy production training and processing large data workloads.

3. Inference Clusters: Production-grade deployment targets used to host trained models as web services for real-time applications.

4. Attached Compute: Existing external Azure compute resources (like Azure Databricks or HDInsight) linked to the workspace for specific tasks.